# RSBEV-Mamba: 3-D BEV Sequence Modeling for Multiview Remote Sensing Scene Segmentation

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***Abstract*— Multiview collaborative perception has been demonstrated to be highly effective in extracting 3-D information from remote sensing scenes by remote sensing bird's-eye-view (RSBEV). However, inherent depth uncertainty in purely visual methods limits view fusion accuracy, and high computational complexity makes it challenging to model long sequences efficiently. To address these issues, we reformulate the BEV segmentation problem as a 3-D sequence modeling task and propose RSBEV-Mamba, a novel framework comprising a 3-D BEV module, a 3-D VMamba module, and a dense BEV contrastive learning module. The 3-D BEV module projects multiview 2-D image features into 3-D world coordinates, thus establishing a foundation for accurate spatial representation. The 3-D VMamba module, based on state-space models (SSMs), optimizes the processing of densely projected features with linear computational complexity in global 3-D spatial modeling. It incorporates a 3-D selective scanning strategy (SS3D) block with 16 scanning strategies, transforming previously ignored projections at different heights into valid 3-D sequences and enriching the contextual depth and precision of BEV encoding. By employing a contrastive learning strategy with the CLIP model, we align BEV and ground truth (GT) features within the same dimensional framework, ensuring spatial integrity after side-view projection. Our approach achieves a 4% improvement mIoU, thus reaching a score of 0.7368 on LEVIR-MDS and surpassing previous state-of-the-art methods. This establishes the 3-D VMamba module as a general model for 3-D perception tasks and sets a new benchmark in remote sensing technology.**

***Index Terms*— Bird's-eye-view (BEV) representation, multiview collaborative segmentation, remote sensing, semantic segmentation.**

## I. Introduction

MULTIVIEW collaborative perception tasks have extensive applications in domains such as multisatellite coordination [2], intelligent transportation [3], [4], drones [5], [6], and robotics [7], [8]. Previous research remote sensing bird's-eye-view (RSBEV) [1] introduced a network tailored for multiview collaborative perception in remote sensing, validated through semantic segmentation tasks. To the authors' knowledge, this is the first implementation of such a task in the remote sensing field. This model processes a set of multiview 2-D images with a common viewing area to predict semantic categories from a BEV on an absolute scale. Initially, the model extracts features from each side view image and projects them onto BEV space, learning terrain category information directly in world coordinates through a transformer-based [9] encoder–decoder structure. While this approach overcomes the limitations of traditional remote sensing perception tasks that only process on the pixel plane and ignore the richness of multiview and 3-D information, several issues still restrict its performance.

First, the inherent lack of depth information in purely visual approaches hinders accurate spatial alignment in BEV modeling. RSBEV [1] employs a multiplane projection method, projecting terrain features at multiple heights and aggregating them with weighted fusion to map the terrain features at ground level. Although this method integrates spatial information into the final BEV feature map, it introduces redundant information, complicating the alignment between BEV features and ground truth (GT) and increasing overfitting risks.

Second, the use of a vision transformer (ViT) [10] as the spatial feature encoder, with its quadratic computational complexity, significantly increases resource demands and limits the scalability of dense segmentation tasks in higher spatial dimensions. Additionally, the lengthy embedding sequence postencoding complicates the learning of contextual information, thus further restricting performance.

Recent advancements in general models have highlighted Mamba [11] as a promising successor to transformers [9] in the field of long sequence prediction, particularly in natural language processing (NLP). This model leverages state-space models (SSMs) for efficient sequence inference with linear complexity. In the domain of natural images, Vim [12] and VMamba [13] have adapted selective scanning modules enabling SSMs to capture contextual information of 2-D image sequences. The integration of SSMs for efficient and powerful context capture into remote sensing tasks has brought forth innovations like RS-Mamba [14] and RSMamba [15], which have been effectively utilized in image classification, semantic segmentation, and change detection [16]. The advent of Voxel Mamba [17] marked a significant advancement in

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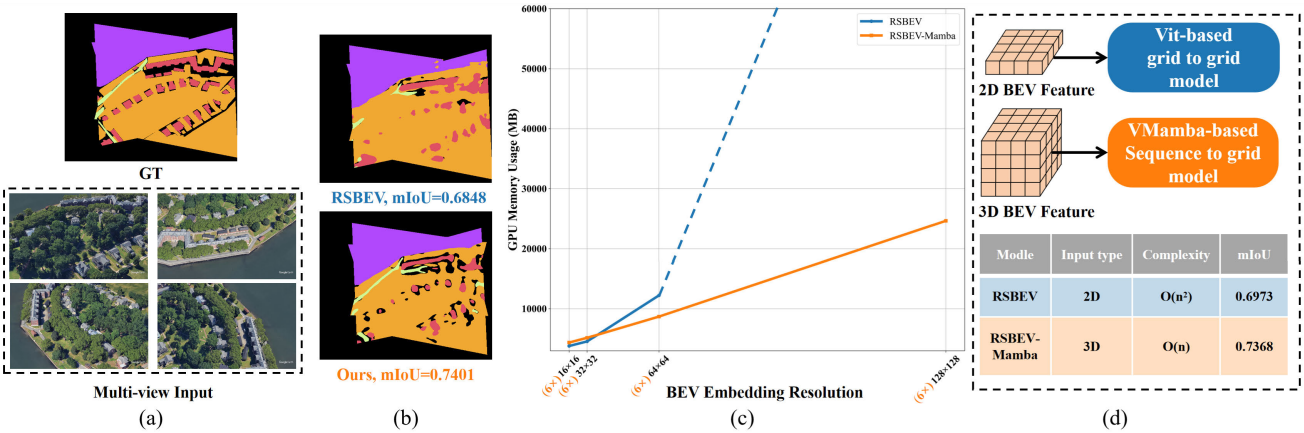



Fig. 1. Comprehensive comparison of performance between RSBEV [1] and our method. (a) Common multiview input images and the GT segmentation map for both models. (b) Prediction results from each model, affirming our model's enhanced capability in accurately integrating global contextual information from complex 3-D scenes. (c) GPU memory usage of both models as the BEV embedding resolution increases, with other parameters held constant. This comparison further validates the linear complexity of our method versus the quadratic complexity of RSBEV. Note that the values in parentheses on the $x$-axis indicate the third dimension in our model. (d) Core differences and performance comparison between the two models, showing input type, complexity, and mIoU scores.

3-D object detection, serving as a robust backbone network for processing point cloud data in 3-D object detection. Voxel Mamba leverages SSMs to manage high-dimensional sequences efficiently, demonstrating its capability in complex 3-D environments. Despite these advancements, the challenge of processing densely arranged sequences in expansive 3-D spaces without compromising on computational efficiency or accuracy remains unmet, highlighting a crucial area for future research in network design and optimization.

Additionally, contrastive learning, exemplified by CLIP [18] in the multimodal domain, has facilitated the association of features across different sources and spaces. In the BEV domain, CLIP-BEVFormer [19] first adopted this paradigm to design an alignment method between targets and features in 3-D object detection networks, ensuring that feature maps possess explicit meanings. Yet, for dense prediction tasks such as semantic segmentation, pixel-by-pixel alignment in the same manner remains challenging.

To address the challenges mentioned, we have rethought the modeling principles of BEV representation, integrating SSMs and contrastive learning techniques to propose RSBEV-Mamba. In the BEV projection phase, our method retains a similar projection approach as in RSBEV, but instead of considering multiple height feature layers as redundant information to be averaged by confidence probabilities, we treat them as a sequence of features in 3-D space. These are directly input into our innovative 3-D VMamba block, designed specifically for 3-D remote sensing. This block extracts global contextual information in space by establishing sequence scanning rules in 3-D spaces. To ensure the spatial accuracy of the projected feature postencoder, we designed a dense BEV contrastive learning module. This module aligns dense BEV features with the GT by calculating the consistency between the GT label map and the average features of each category in BEV, as well as the consistency between the BEV feature map and the GT label map. Finally, we restore and predict terrain categories through an upsampling network structured by FPN [20]. We have compared our method with RSBEV in terms of performance, computational resources, and modeling approach, as shown in Fig. 1.

The contributions of this work can be summarized as follows.

1) *A Novel Framework That Reformulates BEV Segmentation as 3-D Sequence Modeling:* We introduce a new framework that redefines the BEV segmentation problem into a 3-D sequence modeling task. This innovative approach allows the model to process longer sequences across all three spatial dimensions while maintaining linear computational complexity. By efficiently capturing more comprehensive spatial contextual information, our method overcomes the limitations of high computational costs associated with modeling long sequences.
2) *A 3-D VMamba Encoder Module for Perception Tasks:* We have developed the 3-D VMamba module, an innovative encoder that processes sequences of features in 3-D space with a space state model. This encoder significantly advances the field of 3-D perception by allowing for the efficient and accurate extraction of global contextual information from high-dimensional data sequences. This capability is critical for handling the complexities associated with 3-D remote sensing data, providing a substantial improvement over traditional methods that struggle with dense and complex information.
3) *A Dense BEV Contrastive Learning Module:* Our introduction of a dense BEV contrastive learning module marks a significant advancement for dense prediction tasks such as semantic segmentation. This module aligns dense BEV features with GT by evaluating the consistency between different types of feature representations. It ensures the spatial accuracy of the projected features after processing through the encoder, addressing one of the key challenges in maintaining fidelity in feature representation in dense prediction scenarios.

The rest of this article is organized as follows. Section II reviews related work in BEV representation for perception

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tasks, SSMs, and contrastive learning. Section III details our proposed methodology, including preliminaries on RSBEV, the overall architecture, the 3-D VMamba block, the dense BEV contrastive learning module, and the loss function. Section IV provides the experimental results demonstrating the effectiveness and rationale of our method. Finally, we conclude this article and discuss future work in Section V.

## II. Related Work

### A. BEV Representation for Perception Tasks

BEV representation has significantly advanced multisensor collaborative perception. BEV integrates sensor data into a unified aerial perspective through projection transformations, rather than simply merging results from each sensor [21]. This integration facilitates the use of tailored backbone networks for specific downstream tasks. Presently, prominent methods for constructing BEV representations include employing inverse perspective mapping (IPM) to explicitly form relationships [22], [23], and using complex mapping functions to implicitly establish connections between side views and the aerial view [24], [25].

In explicit BEV methods, depth-based transformations in autonomous driving avoid plane assumption errors by integrating depth data through LiDAR [26], [27]. Methods generating pseudo-LiDAR from depth estimates add significant inference load and can degrade performance in perception tasks [28]. These systems typically utilize convolution-based networks or self-attention mechanisms, ensuring that the feature maps maintain explicit meanings throughout the process.

Implicit methods started with MLPs [29], [30] to learn mappings from camera views to BEVs without relying on direct camera parameter projections. With the evolution of transformer-based techniques, methods employing cross-attention mechanisms have been developed to directly convert contextual information from camera views into BEV feature maps [25], [31], [32]. These are particularly effective for sparse perception tasks like 3-D object detection, which can utilize sparse queries [24], [33], [34]. For denser perception tasks such as semantic segmentation and change detection, ensuring explicit spatial definitions of feature maps is crucial, often achieved through cross-attention querying of each BEV grid [25], [35], [36]. This approach, while effective for various downstream tasks, leads to increased computational complexity. It is also important to note that in the cross-attention mechanism, the keys and queries originate from multiview image features and BEV spatial embeddings, respectively. The direct application of these unaligned features makes it challenging to ensure the spatial accuracy of the output BEV feature maps [28].

However, remote sensing images, taken from high altitudes, inherently focus on broader global information but often fail to implicitly capture the detailed spatial context that is more apparent in natural images taken at ground level. Therefore, in multiview collaborative perception tasks, RSBEV [1] chooses to explicitly represent BEV features to address the significant variance in depth changes found in remote sensing, as opposed to the automotive domain. This method involves integrating a series of depth features, which are then selectively filtered within the encoder using a self-attention mechanism. Despite these advances, there is still a scarcity of research on effectively leveraging spatial information between multiple viewpoints with uncertain poses, highlighting a key area for future exploration.

### B. State-Space Models

While transformer-based models are highly perceptive, their quadratic complexity makes modeling long-sequence correlations challenging. Inspired by control theory, some studies [37], [38], [39] have utilized the discretization of continuous state space to construct complex temporal dynamics, proving impactful in deep learning. The LSSL method [38] under HiPPO [39] initialization has shown potential to match CNNs and transformers in handling long-term dependencies though it still falls short in computational efficiency. Subsequently, the S4 model [40] introduced a normalization strategy with diagonalized parameter matrices, significantly reducing computational and memory costs. Building on this, the S5 model [41] achieved multi-input multi-output (MIMO) processing, and Mamba [11] incorporated a selection mechanism, enhancing these models' capabilities as general backbone networks.

In the realm of visual processing, substantial strides have been made with the advent of Mamba-based models, particularly in handling 2-D image sequences. The Vim model [12] introduces an innovative approach by flattening images into 2-D planes and employing horizontal selective scanning, which enables enhanced global interaction across image segments. In a similar vein, VMamba [13] delves deeper into capturing significant contextual correlations by scanning both horizontally and vertically, utilizing SSMs to process each direction independently. These strategies signify a pivotal advancement in the efficient modeling of the global context in images, showcasing the potential of selective scanning techniques to vastly improve the understanding and interpretation of 2-D visual sequences in diverse applications. Additionally, RS-Mamba [14] adapts to the isotropic characteristics of remote sensing images with an eight-directional scanning strategy, thus enhancing global information perception. In the 3-D perception area, the unique properties of point cloud data have propelled methods like [17] to state-of-the-art status for sparse 3-D representations, significantly advancing the handling and analysis of 3-D spatial data. Similarly, MambaBEV [42] designed a state-space backbone to efficiently fuse temporal information in BEV representations, specifically targeting 3-D object detection in autonomous driving tasks.

In dense prediction tasks, the dense arrangement of 3-D spatial features in the depth dimension presents challenges for existing SSMs to capture contextual information across all three dimensions simultaneously. Therefore, further exploration is needed for a general 3-D vision Mamba model to effectively handle these complexities.

### C. Contrastive Learning

Contrastive learning, originating from cognitive science and statistics, has evolved into a cornerstone of unsupervised

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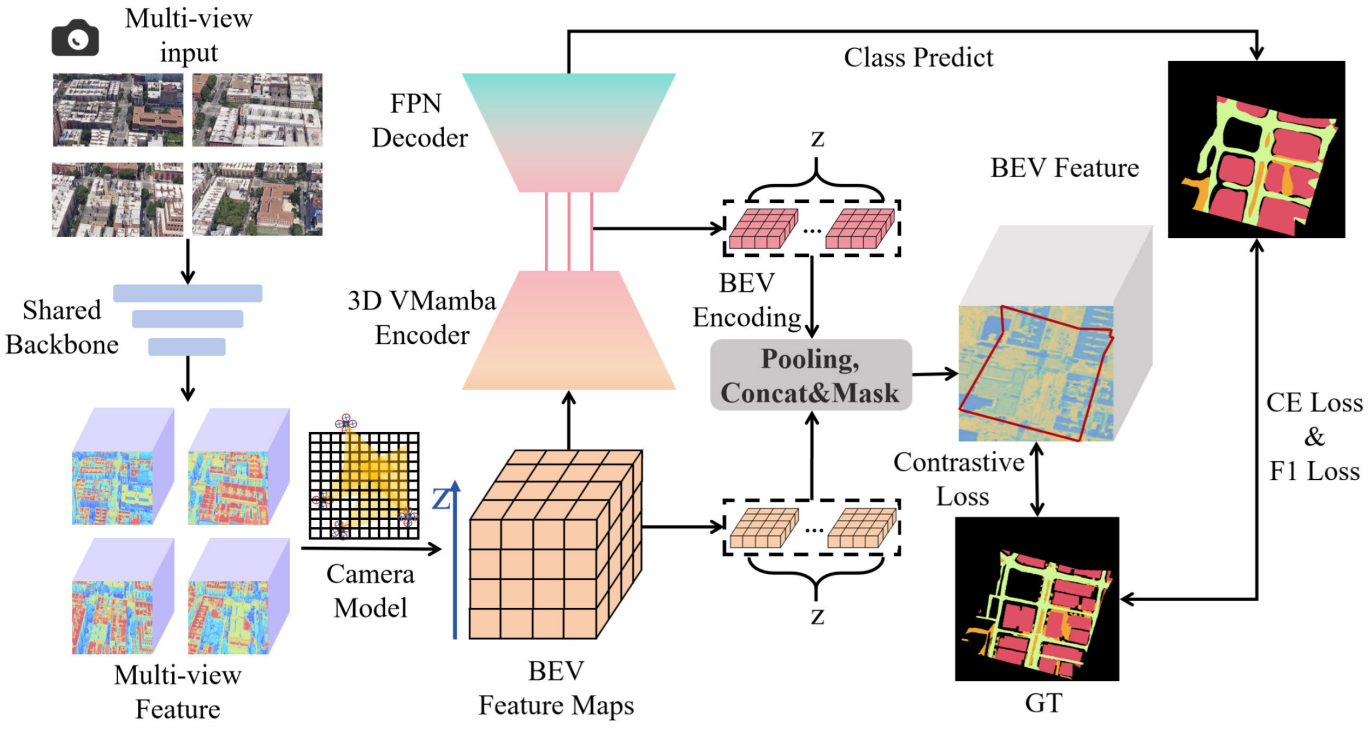



Fig. 2. Process from multiview input to semantic segmentation. The multiview images are passed through a shared backbone and then projected into BEV feature space using the camera model. A 3-D VMamba encoder–decoder structure is used for class prediction. Additionally, a dense BEV contrastive learning branch aligns the concatenation of embeddings and encoded features with the GT to ensure the explicit meaning of projected features.

learning in computer vision. The CLIP model [18] showcases this by efficiently learning visual concepts through natural language supervision. It predicts the most accurate caption for a given image from a set of possibilities, demonstrating strong transferability to various downstream tasks without task-specific training. CLIP also illustrates how contrastive learning can capture high-level semantic information across different modalities and data spaces.

In the domain of semantic segmentation, Lseg [41] utilizes a method similar to CLIP by using class prompts as textual inputs. This approach calculates similarities to achieve zero-shot semantic segmentation, effectively replacing true labels with textual labels and thus necessitating additional text annotations for each image. Meanwhile, GroupViT [43] introduces a grouping block and learnable group tokens, employing a clustering-like method for unsupervised segmentation. This technique functions more as an image encoder, thus limiting its effectiveness in dense prediction tasks. In the multiview fusion perception domain, CLIP-BEVFormer [19] significantly enhances 3-D object detection tasks by integrating the GT-BEV module. This involves using CLIP to align GT and encoded features during the BEV encoding stage, ensuring explicit significance in the feature maps. While this method markedly improves performance over traditional approaches [25], it primarily enhances explicitness in targeted areas within the feature maps, presenting challenges for pixel-by-pixel alignment in dense prediction tasks.

To address the challenges, we introduce a dense BEV contrastive learning module, which divides the objective function into two parts. It balances the similarity between each class's true label and its corresponding feature, as well as the similarity between all context-enriched BEV spatial features and their respective labels. This method ensures the spatial accuracy of the projected features after processing through the encoder, thus enhancing the fidelity and effectiveness of the BEV representation in multiview perception tasks.

## III. Proposed Method

### A. Preliminaries: RSBEV

This section outlines the foundational workflow and theoretical underpinnings of the RSBEV model, inspired by robust multisensor fusion capabilities observed in autonomous driving. The task involves processing a set of multiview images from a 3-D remote sensing scene to obtain BEV scene information within a common view area in the world coordinate system. The task is structured into three main components: the BEV Feature Builder, the Spatial Feature Encoder, and the FPN-Based Decoder, which is utilized multiple times throughout the process.

We first describe the modeling process for BEV, which projects 2-D images from various perspective views $\mathbf{I} = \{\mathbf{I}_1, \mathbf{I}_2, \ldots, \mathbf{I}_N\}$ using corresponding camera parameters $\mathbf{M} = \{\mathbf{M}_1, \mathbf{M}_2, \ldots, \mathbf{M}_N\}$ into a BEV feature map $\mathbf{B}$.

Initially, an image feature extraction network, denoted as NET, processes the multiview input images $\mathbf{I}$ to compute the image features. Additionally, a mask weight, **mask**, is applied elementwise to the features extracted by NET. This operation effectively scales the features based on their relevance or visibility in the particular camera view. Each image feature is associated with coordinates on pixel plane, which are then

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projected back to the 3-D world coordinates corresponding to a specific grid point $(x, y, z)$ in the BEV grid. The transformation $M_{\text{trans}}(\mathbf{M}_i, x, y, z)$ describes how a 3-D point $(x, y, z)$ in the world coordinate system projects onto a 2-D pixel plane through the intrinsic and extrinsic camera parameters contained within matrix $\mathbf{M}_i$

$$\begin{pmatrix} u_i \\ v_i \\ 1 \end{pmatrix} = \mathbf{K}_i \begin{pmatrix} \mathbf{R}_i & \mathbf{t}_i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}. \tag{1}$$

Here, $(u_i, v_i)$ represents the feature pixel coordinates from the $i$th viewpoint, with $\mathbf{K}_i$ being the intrinsic matrix, $\mathbf{R}_i$ the rotation matrix, and $\mathbf{t}_i$ the translation vector, collectively forming the extrinsic matrix. The transformation matrix $\mathbf{M}_i$ for each camera is, thus, constructed as the product of $\mathbf{K}_i$ and the joined extrinsic matrix $\begin{pmatrix} \mathbf{R}_i & \mathbf{t}_i \end{pmatrix}$.

The resulting features in $(x, y, z)$ are represented by $\mathbf{F}_{\text{3-D}}(x, y, z) \in \mathbb{R}^{N \times C}$, where $N$ is the number of cameras and $C$ is the dimension of image features. This matrix represents the collection of image features from the image planes, whose coordinates are obtained by projecting the BEV grid point $(x, y, z)$ onto the corresponding image planes. For positions within the BEV grid that are outside the projected image feature region, the corresponding entries in $\mathbf{F}_{\text{3-D}}$ are filled with zero vectors $\mathbf{0} \in \mathbb{R}^{1 \times C}$. This method ensures that each grid point in the BEV representation retains spatially accurate and relevant image features, facilitating accurate scene understanding and analysis. The BEV space is partitioned into multiple layers at predefined heights specified by the set $\mathbf{z} = \{z_1, z_2, \ldots, z_L, \ldots, z_{2L-2}, z_{2L-1}\}$. For each layer corresponding to a specific height, a grid-shaped BEV map is established. Image features within each grid point $\mathbf{F}_{\text{3-D}}(x, y, z_i)$ are obtained using the method described previously. Then, they undergo a fusion process as delineated in the following equation:

$$\begin{aligned} \mathbf{B}_i(x, y) &= \frac{1}{n_{\text{hit}}} \sum_{n=1}^{N} \mathbf{F}_{\text{3-D}}(x, y, z_i)[n], \text{ where} \\ n_{\text{hit}} &= \max(1, |\{n : \mathbf{F}_{\text{3-D}}(x, y, z_i)[n] \neq \mathbf{0}\}|). \end{aligned} \tag{2}$$

Here, $\mathbf{B}_i \in \mathbb{R}^{\text{grid_h} \times \text{grid_w} \times C}$ represents the BEV map of height layer $i$. Finally, weighted pooling is employed to aggregate $\mathbf{B}_i$ according to the discrete Gaussian distribution probabilities, thus yielding the final BEV feature map $\mathbf{B}$

$$\mathbf{B} = \sum_{i=0}^{2L-1} \frac{1}{\sigma\sqrt{2\pi}} \exp - \frac{(z_i - z_L)^2}{2\sigma^2} \cdot \mathbf{B}_i. \tag{3}$$

It is crucial to recognize that, due to the adoption of the ViT [10] architecture within the subsequent BEV space encoder—which exhibits quadratic complexity—the sequence length limitation encourages RSBEV to implement probabilistic prior-weighted pooling of BEV space features into 2-D features. These features are subsequently refined through a self-attention mechanism to extract spatial information. The spatial feature encoder processes the input $\mathbf{B}$, dividing it into sequence patches for embedding, which are then transformed into input tokens ($\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$) through a linear projection. This setup employs the self-attention mechanism to acquire the BEV encoding, where one attention head can be articulated as

$$\text{SelfAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V}. \tag{4}$$

Finally, an FPN [20] is introduced within the segmentation output head. By employing both upscale UP modules and downscale DOWN modules, we fuse the BEV map from prior to the spatial feature encoder with the feature maps concatenated with BEV patch encoding. This achieves multiscale fusion, ultimately yielding category segmentation results in the BEV space at the original scale.

### B. Overall Architecture

In this work, we focus on multiview collaborative semantic segmentation tasks, specifically aiming to obtain BEV semantic segmentation results in the world coordinate system from a set of multiview images of a 3-D remote sensing scene within the common view area. Building upon RSBEV, we have developed a new framework named RSBEV-Mamba. The overall workflow is depicted in Fig. 2.

We first extract features from the input multiview images using a shared backbone network, where a ResNet-50 [44] architecture is employed. However, rather than directly embedding in pixel space, we explicitly project the features into BEV space. This projection allows us to capture more accurate spatial relationships between different views, addressing the challenge posed by the variability of viewpoints in drone-based remote sensing. Following this projection, we transform the task into a 3-D sequence modeling problem to ensure the semantic information from different viewpoints is correctly mapped to the BEV space. The final pooling and fully connected layers of the ResNet-50 backbone are replaced with an FPN [20] upsampling network, which ensures that the output image feature maps are half the size of the input image dimensions. This approach improves the representation of the scene, ensuring accurate semantic mapping even with varying drone viewpoints.

In the BEV modeling part, considering the variability and high variance in ground elevations from drone perspectives, we utilize the camera model to explicitly project 2-D image features into 3-D BEV space. Unlike the approach in RSBEV where multiple height feature layers were considered redundant and filtered out to determine the correct features at each grid, in RSBEV-Mamba, we treat each elevation as providing valuable contextual information for global perception. Therefore, we regard these layers as sequences of features in 3-D space. This methodology is also facilitated by our design of the 3-D VMamba Module, which can handle sequences of length comparable to the quadratic complexity of VIT under the same computational complexity.

We employ a segmentation model structured around an encoder–decoder architecture, where the encoder is constituted by our 3-D VMamba module, and the FPN [20] is integrated across both the encoder and decoder to enhance feature resolution at various scales. This design allows the 3-D VMamba module to handle complex encoding of spatial

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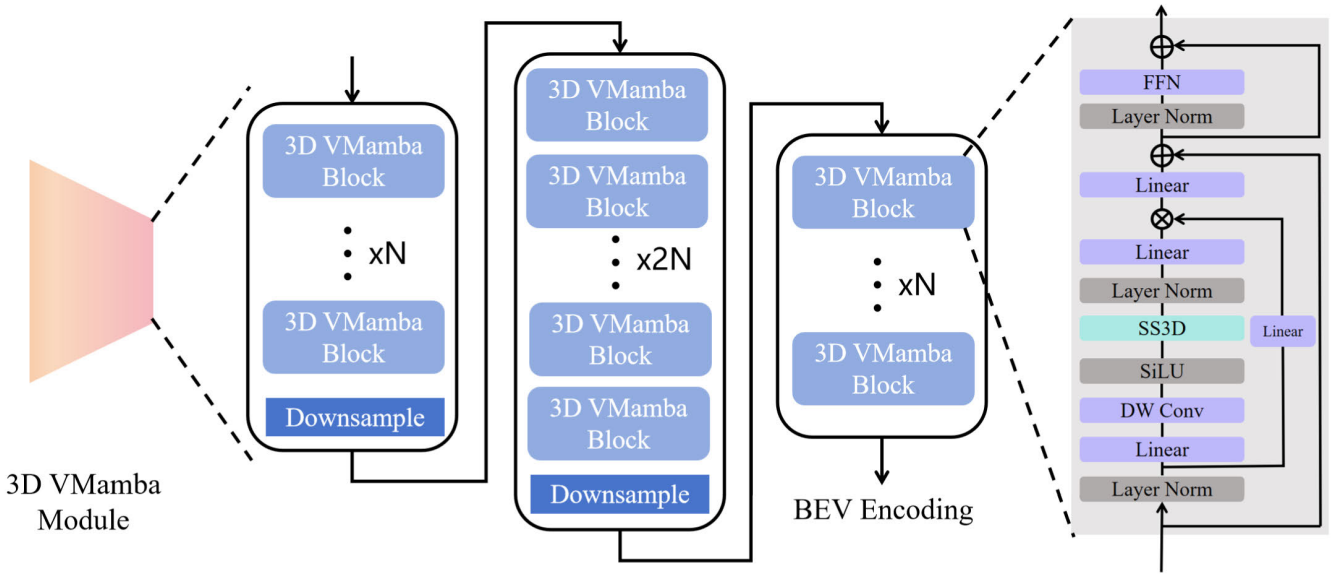



Fig. 3. Structure of the 3-D VMamba-based encoder, which consists of three stages with modules scaled in a spindle-shaped configuration. Each stage is equipped with a varying number of 3-D VMamba blocks: $N$(we set 3) blocks at both the initial and final stages, and $2N$ blocks in the middle stages. Except for the last stage, every stage involves a downsampling process that doubles the depth dimension while keeping the height dimension constant. A key innovative component within the 3-D VMamba block is the SS3D.

features effectively, while the FPN enhances segmentation capabilities by facilitating detailed multiscale feature integration across the decoder stages. Initially, the BEV space features $\mathbf{B} \in \mathbb{R}^{n_z \times \text{bev_h} \times \text{bev_w} \times C}$ are transformed into a 3-D feature embedding sequence $\mathbf{F}_{\text{embedding}} \in \mathbb{R}^{n_z \times (\text{bev_h}/p) \times (\text{bev_w}/p) \times d}$, where $n_z$ is the number of layers in the height dimension, $p$ is the patch size, and $d$ is the dimension of the embeddings. The embeddings are channeled through multiple encoder stages, each furnished with a varying number of 3-D VMamba blocks arranged in a spindle-shaped configuration—fewer at the beginning and end stages, and more concentrated in the middle stages. Except for the final stage, each encoder stage engages in a downsampling process where the depth dimension is doubled while the height dimension remains constant. This method preserves vital vertical details while amplifying the capacity to discern finer spatial attributes through enhanced depth resolution. The decoder stage integrates features through skip connections and multiscale fusion, and height layers are eliminated through pooling at a scale of twice the patch embedding, with subsequent upsampling-convolution operations restoring the original scale for category prediction.

Furthermore, during training, we introduce an additional branch, the dense BEV contrastive learning module. After height layers are eliminated through pooling, we engage in contrastive learning between the feature vectors and the linearly projected GT map features to further align the BEV feature map, ensuring its accurate explicit representation.

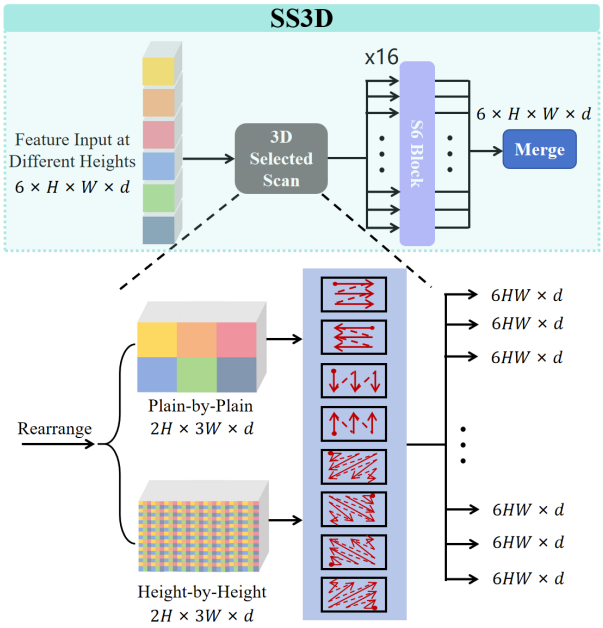



Fig. 4. Architecture of the SS3D in 3-D VMamba block. Initially, 3-D features are flattened into two types of 2-D feature maps: one organized by plane and another by height. Each layout undergoes a scan across horizontal, vertical, diagonal, antidiagonal, and their respective flipped directions, producing eight unique feature sequences per arrangement, totaling 16. These sequences are then fed into an S6 block, and the outputs are merged in sequence order to enhance feature representation and integration.

### C. 3-D VMamba Block

To model the BEV features across multiple elevations more effectively, we adopt the SSM as the foundation for sequence modeling. SSMs are particularly suited for tasks involving spatially structured data, such as 3-D BEV features, due to their linear computational complexity and ability to efficiently capture long-range dependencies. In contrast, RNN-based architectures, such as LSTM [45] and seq2seq [46], are better suited for sequential data in NLP but struggle with the scalability required for high-dimensional spatial data. Moreover, while Transformer-based models have demonstrated strong capabilities in modeling spatial dependencies, their

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quadratic complexity makes them less efficient for dense tasks like BEV segmentation.

Our approach leverages the strengths of SSMs through the 3-D VMamba block, which incorporates a 3-D selective scanning strategy (SS3D) for efficient processing of global spatial information. As illustrated in the rightmost part of Fig. 3, the encoder is composed of multiple stages, each stacked with 3-D VMamba blocks. Building upon the architecture of RSMamba [14], we introduce an innovative shortcut operation after the first layer norm in each block. This operation involves elementwise multiplication of linearly projected vectors, enhancing feature integration across different elevations.

The SS3D module, as shown in Fig. 4, selectively scans along spatially significant directions, enabling the block to effectively capture global context across 3-D sequences. This design ensures meaningful contributions from features at various elevations to the BEV representation, addressing challenges like redundant or misaligned information in prior methods. To integrate the height dimension information into the spatial dimensions, 3-D features $\mathbf{F} \in \mathbb{R}^{n_z \times H \times W \times d}$, with $n_z$ set to 6, are rearranged into two types of 2-D feature maps: one organized by plane and another by height. These arrangements are represented as $\mathbf{F}_{\text{2-D_plane}} \in \mathbb{R}^{2H \times 3W \times d}$ and $\mathbf{F}_{\text{2-D_height}} \in \mathbb{R}^{2H \times 3W \times d}$, respectively.

In designing the flattening strategy, we considered the unique characteristics of BEV space. Strong semantic correlations exist within each layer along all directions and across the same position at different heights. Thus, we adopt two complementary flattening methods: plane-by-plane flattening emphasizes semantic information within each $xy$ plane, while height-by-height flattening focuses on semantic relationships across different heights for the same spatial position. This dual strategy allows the model to capture comprehensive contextual information.

After rearranging the features, each map undergoes scanning in multiple directions, including horizontal, vertical, diagonal, antidiagonal, and their respective flipped counterparts. This scanning process generates eight unique feature sequences per arrangement, producing a total of 16 sequences. Each sequence $\mathbf{s}$ is represented as $\mathbf{s} \in \mathbb{R}^{6HW \times d}$ and processed through an S6 block. These processed sequences are then rearranged back into their original 3-D layout and summed to merge the sequence information

$$\textbf{SS3D_output} = \sum_{i=1}^{16} \text{rearrange}(\text{S6}(\mathbf{s}_i)) \tag{5}$$

where $\textbf{SS3D_output} \in \mathbb{R}^{6 \times H \times W \times d}$ represents the unified output enriched with spatial and height contextual details. This methodology effectively integrates features from multiple perspectives into a cohesive 3-D representation, enabling the model to process long sequences in 3-D space with enriched semantic information and minimal redundancy.

### D. Dense BEV Contrastive Learning Module

After modeling the global contextual information with the 3-D VMamba module and obtaining the downsampled BEV feature encoding, there remains a lack of evidence to ensure that the explicitly modeled BEV feature space is aligned with real-world terrain coordinates. To address this, our training incorporates an additional branch that employs contrastive learning to enhance alignment between the two. This module diverges from methods like CLIP-BEVFormer [19] that align sparse targets directly with BEV features. Instead, it is designed for the intricacies of dense prediction tasks like segmentation, where direct alignment for each grid cell is computationally prohibitive and could dilute essential contextual information. This module is designed to balance positional accuracy with feature interrelations, optimizing for dense tasks. The pseudocode detailing these operations is shown in Algorithm1.

**Algorithm 1** Dense BEV Contrastive Learning Module

**Input:** Ground truth $\mathbf{G}$, BEV encoding $\mathbf{E}$, Gradient mask $\mathbf{M}$
**Input:** Learnable parameters $\lambda_{\text{cls}}$ and $\lambda_{\text{gt}}$
**Output:** Cross-entropy losses $\text{L}_{\text{cls}} + \text{L}_{\text{gt}}$

**Step1: Compute BEV averages and flattened features:**
  $\mathbf{A}, \mathbf{F}, \mathbf{G_f} \leftarrow$ `ComputeFeatures`$(\mathbf{E}, \mathbf{G}, \mathbf{M})$
  // $\mathbf{A}$: BEV average features, $\mathbf{F}$: Flattened BEV features
  // $\mathbf{G_f}$: Flattened GT indexes
**Step2: Process GT and project features:**
  $\mathbf{G_p}, \mathbf{A_p}, \mathbf{F_p} \leftarrow$ `Proj(GetMaps`$(\mathbf{G}), \mathbf{A}, \mathbf{F})$
  $\mathbf{A_n}, \mathbf{G_n}, \mathbf{F_n} \leftarrow$ `Norm`$(\mathbf{A_p}, \mathbf{G_p}, \mathbf{F_p})$
**Step3: Compute similarity scores:**
  $\mathbf{S}_{\text{gt}} \leftarrow \mathbf{A_n} \times \mathbf{G_n}^{\top}$
  // $\mathbf{S}_{\text{gt}}$: BEV averages and GT feature
  $\mathbf{S}_{\text{cls}} \leftarrow \mathbf{F_n} \times \mathbf{G_f}^{\top}$
  // $\mathbf{S}_{\text{cls}}$: Flattened features and GT feature
**Step4: Apply scaling and softmax:**
  $\mathbf{P}_{\text{gt}}, \mathbf{P}_{\text{cls}} \leftarrow$ `Softmax`$(\mathbf{S}_{\text{gt}} \times \exp(\lambda_{\text{gt}}), \mathbf{S}_{\text{cls}} \times \exp(\lambda_{\text{cls}}))$
**Step5: Calculate Loss for alignment and classification:**
  $\text{L}_{\text{gt}} \leftarrow$ `CrossEntropy`$(\mathbf{P}_{\text{gt}}, \text{Labels})$
  $\text{L}_{\text{cls}} \leftarrow$ `CrossEntropy`$(\mathbf{P}_{\text{cls}}, \mathbf{G_f})$
**return** $\text{L}_{\text{gt}} + \text{L}_{\text{cls}}$

First, the GT map is resized and rerepresented as a Boolean tensor $\textbf{gt} \in \mathbb{R}^{n_{\text{cls}} \times 224 \times 224}$, where $n_{\text{cls}}$ represents the total number of classes. Then, this tensor is input into a pretrained image encoding model, RN101 [18], which transforms the GT into CLIP features for each class. Notably, the RN101 model remains frozen during training, ensuring no gradient updates are applied. The transformation is described as

$$\textbf{gt_clip} = \text{RN101}(\textbf{gt}) \tag{6}$$

where $\textbf{gt_clip} \in \mathbb{R}^{n_{\text{cls}} \times 512}$ represents the CLIP feature embeddings of the GT for each class.

For the BEV features, two operations are performed in parallel. First, the BEV features corresponding to each class position in the GT map are averaged to produce $\textbf{bev_avgs} \in \mathbb{R}^{n_{\text{cls}} \times d}$, where $d$ denotes the feature dimension. Second, all features within the GT map range are flattened to form $\textbf{bev_flatten} \in \mathbb{R}^{n_{\text{grad}} \times d}$, accompanied by the corresponding GT labels $\textbf{gt_flatten} \in \mathbb{R}^{n_{\text{grad}} \times 1}$, where $n_{\text{grad}}$ is the total number of features contributing to the gradient computation.

Subsequently, all feature vectors, including **bev_avgs**, **bev_flatten**, and **gt_clip**, are projected into the same

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low-dimensional embedding space (64-D). Following this projection, two cosine similarity calculations are performed to align the BEV features with the GT. The first similarity score compares **bev_avgs** with **gt_clip**, emphasizing the aggregate alignment across the entire scene. This component (GT loss) ensures that the BEV features align spatially and semantically with the GT terrain. The second similarity score compares **bev_flatten** with **gt_clip**, focusing on precise alignment at specific feature locations. This component (CLS loss) enhances classification accuracy for individual grid cells in the BEV map.

These similarity scores are scaled using learnable scaling factors $\lambda_{\text{gt}}$ and $\lambda_{\text{cls}}$, normalized using the softmax function, and finally evaluated using cross-entropy loss. The combined loss balances global spatial alignment (via GT loss) and local feature classification accuracy (via CLS loss), optimizing both the spatial fidelity and the semantic precision of the BEV representation.

Following the projection and similarity score computation, learnable scaling factors $\lambda_{\text{gt}}$ and $\lambda_{\text{cls}}$ are applied to the similarity scores $S_{\text{gt}}$ and $S_{\text{cls}}$, respectively. Once the scores are scaled, they are normalized using the softmax function to produce $\mathbf{P}_{\text{gt}}$ and $\mathbf{P}_{\text{cls}}$, which represent the final probability distributions.

Finally, cross-entropy loss is calculated for both GT alignment and classification to ensure accurate feature alignment and classification. This results in two loss values: $L_{\text{gt}}$ for GT alignment and $L_{\text{cls}}$ for classification.

### E. Loss Function

Our loss function comprises the weighted cross-entropy loss $L_{\text{ce}}$, Dice loss [47] $L_{\text{dice}}$, and the contrastive loss $L_{\text{con}}$ from the dense BEV contrastive learning module. The definitions of $L_{\text{ce}}$ and $L_{\text{dice}}$ remain consistent with those used in RSBEV [8], while $L_{\text{con}}$ is composed of $L_{\text{cls}}$ and $L_{\text{gt}}$, which have been detailed in Section III-D. The loss for GT alignment, $L_{\text{gt}}$, is computed using the cross-entropy function as follows:

$$L_{\text{gt}} = \text{CrossEntropy}\left(\mathbf{P}_{\text{gt}}, \text{Labels}\right) \tag{7}$$

where $\mathbf{P}_{\text{gt}} \in \mathbb{R}^{n_{\text{cls}} \times n_{\text{cls}}}$ represents the predicted probability distribution for the GT alignment, where $n_{\text{cls}}$ is the number of classes. The GT labels, denoted as Labels $\in \mathbb{R}^{n_{\text{cls}} \times 1}$, are the true class labels corresponding to each sample in the BEV space.

Similarly, the classification loss, $L_{\text{cls}}$, is calculated with

$$L_{\text{cls}} = \text{CrossEntropy}(\mathbf{P}_{\text{cls}}, \mathbf{G}_{\mathbf{f}}) \tag{8}$$

where $\mathbf{P}_{\text{cls}} \in \mathbb{R}^{n_{\text{grad}} \times n_{\text{cls}}}$ is the predicted probability distribution for the flattened BEV features, where $n_{\text{grad}}$ is the number of features involved in gradient computation. $\mathbf{G}_{\mathbf{f}} \in \mathbb{R}^{n_{\text{grad}} \times 1}$ represents the GT class indices for each feature in the flattened BEV feature map.

The total loss is the sum of these two components

$$L_{\text{contra}} = L_{\text{gt}} + L_{\text{cls}}. \tag{9}$$

Finally, the overall loss $L$ can be expressed as

$$L = L_{\text{ce}} + L_{\text{dice}} + \lambda \cdot L_{\text{contra}} \tag{10}$$

where $\lambda$ is set to 0.3, as determined through subsequent experiments, achieving the optimal balance between high performance and fast convergence.

## IV. Experiment

### A. Dataset

We conducted a comprehensive comparison of our method with a state-of-the-art approach and other leading methods from autonomous driving on the multidrone collaborative segmentation dataset, *LEVIR-MDS* [1]. This dataset consists of ten community-level scenes across three continents, each containing time-sequenced frames captured from four drone perspectives (1600 × 900 resolution), with over 5800 frames/scene. It leverages real-world 3-D scenes generated from Google Earth Studio, with annotations for six categories: buildings, roads, vegetation, water, ships, and others. We used eight scenes for training, one for validation, and one for testing.

### B. Experimental Settings

*1) Implementation Details:* In our experiments, the BEV features are accommodated in a tensor with dimensions of 512 × 512 × 6, corresponding to a spatial resolution of 0.8 × 0.8 × 5 m, effectively covering a perception range of 409.6 × 409.6 × 30 m. We deployed RSBEV-Mamba using PyTorch [48] on a single NVIDIA GeForce RTX 4090 24-GB GPU. For the learning rate configuration during training, we set the learning rate to $1 \times 10^{-5}$ for the pretrained ResNet-50 backbone, $1 \times 10^{-3}$ for the 3-D VMamba blocks in the encoder, and $5 \times 10^{-4}$ for other learnable parameters such as projections and convolutions. The Adam optimizer [48] was employed with a weight decay of $1 \times 10^{-7}$ and a learning rate decay of 0.95 per epoch. The model was trained for over 100 epochs to ensure full convergence.

*2) Evaluation Metrics:* To evaluate the model's performance, we measure the intersection over union (IoU) for each class, the mean IoU (mIoU) across all classes, and the average F1-score per class. IoU quantifies the accuracy of the segmentation results by calculating the overlap between the predicted and GT areas. The F1-score balances precision and recall by computing their harmonic mean, thus offering a comprehensive evaluation of the model's performance.

*3) Data Augmentation:* We adopt the data augmentation methods inspired by [23], applying augmentations both in the pixel space before the shared backbone and in the BEV space before the encoder. The former focuses on enhancing image color, including brightness variations (p = 0.5). The latter concentrates on spatial augmentations, including random flipping (horizontal: p = 0.5 and vertical: p = 0.5), scaling (0.8–1.2), and rotation (−90° to 90°). Additionally, we should also apply the same spatial transformation to the corresponding BEV GT map and the mask that records portion involved in gradient computation. Fig. 5 illustrates a comparison of our model's performance with and without data augmentation during training and validation.

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE VALIDATION OF THE LEVIR-MDS [1] DATASET

| Modeling Approach | Model | IoU_vegetation | IoU_building | IoU_road | IoU_ship | IoU_water | mIoU↑ | F1 ↑ |
|---|---|---|---|---|---|---|---|---|
| Implicit | BEVFormer[25] | 0.1516 | 0.0437 | 0.0964 | 0.0252 | 0.1235 | 0.1184 | 0.0707 |
| Explicit | LSS[22] | 0.3514 | 0.4793 | 0.5102 | 0.7639 | 0.5770 | 0.5969 | 0.3932 |
| Explicit | BEVDet[23] | 0.3268 | 0.5077 | 0.4821 | 0.7728 | 0.7041 | 0.6143 | 0.4180 |
| Explicit | RSBEV[1] | 0.5931 | **0.6536** | 0.5077 | 0.8324 | 0.8362 | 0.6973 | 0.5910 |
| Explicit | Ours | **0.6367** | 0.6522 | **0.5668** | **0.8776** | **0.8656** | **0.7368** | **0.6232** |

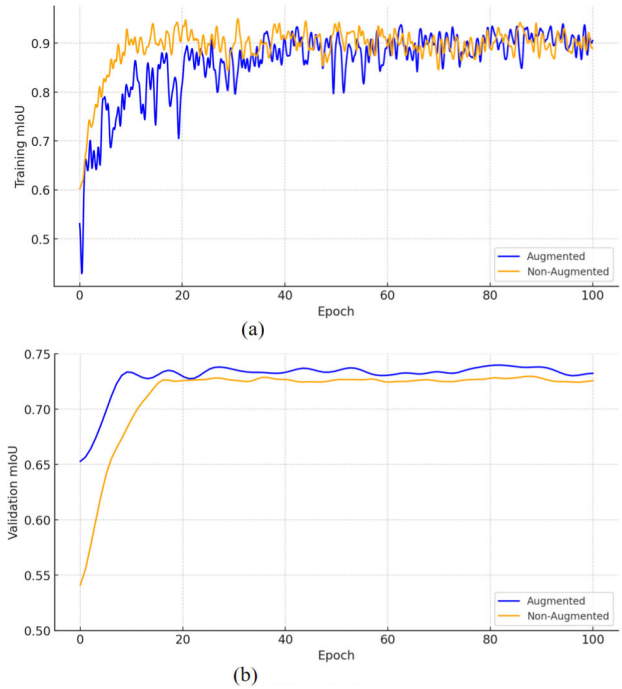



Fig. 5. Comparison of mIoU curves for training and validation with (blue) and without (orange) data augmentation. The validation plot highlights an increase in overall mIoU, with the model achieving 74.89% mIoU with augmentation compared to 73.96% without, marking an improvement of 0.93%. (a) Training mIoU. (b) Validation mIoU.

### C. Comparison With the State-of-the-Art Methods

We compared our method with *BEVFormer* [25], *LSS* [22], *BEVDet* [23], and *RSBEV* [1]. Among them, LSS and BEVDet are representative methods for explicit BEV representation modeling using camera poses and depth in autonomous driving tasks, while BEVFormer represents implicit BEV representation through cross-attention. In our experiments, we modified the inputs of these frameworks to accept multiview inputs from arbitrary poses and replaced the output heads of [23] and [25] with FPN-based segmentation networks to generate segmentation results. The comparison of segmentation results is shown in Fig. 6. Meanwhile, Table I presents the performance comparison of different models on the LEVIR-MDS validation set. It is clear that our model achieved the highest mIoU (0.7368) and F1-score (0.6232) across all methods. From both the data in Table I and the visual comparison, it can be observed that our method demonstrates significant accuracy improvements in objects such as roads and ships, which have fewer pixels and more complex texture information (IoU for roads increased by 6% and for ships by 4.5%).

In addition to the segmentation performance, we also compared the inference speed (frames/s) of the methods. The segmentation performance (mIoU) versus inference speed (frames/s) of five methods is shown in Fig. 7, where *LSS* and *BEVDet* are CNN-based architectures, *BEVFormer* is a Transformer-based model, and *RSBEV* and *Ours* are models that incorporate advanced BEV processing techniques. The comparison shows that, while our method achieves the highest performance, it also maintains competitive inference speed, balancing accuracy with efficiency.

It is worth noting that BEVFormer [25] achieved an mIoU of only 0.1184 in our task, which is close to the theoretical expected value for a random distribution, 0.0909. Therefore, this method is ineffective for our task. We hypothesize that BEVFormer, which employs implicit modeling, relies heavily on the input data and primarily distinguishes multiview inputs through positional encoding. However, in our task, the relative positions of the multiview inputs are inherently nonfixed, making it difficult for the cross-attention mechanism in implicit modeling to learn the mapping from pixel space to BEV space with limited data.

Additionally, to further validate the collaborative capability to multiview inputs, we divided the common visible area of each frame's multiview input into four regions based on the number of overlapping viewpoints (ranging from 1 to 4, as shown in Fig. 8. For each of these regions, we calculated the IoU and F1-score, with the results presented in Table II. The results indicate that as the number of overlapping viewpoints increases, the detection accuracy improves accordingly. The data clearly show that as the number of overlapping viewpoints increases, segmentation accuracy correspondingly improves. This trend is especially pronounced in regions prone to occlusion, such as vegetation and roads, demonstrating the robustness of our method in handling complex multiview scenarios.

### D. Ablation Study

To validate the effectiveness of different components of our method, we conducted a series of experiments on the LEVIR-MDS dataset. We compared the 3-D VMamba block and the dense BEV contrastive learning module with their respective variants.

For the 3-D VMamba block, we conducted the following comparisons: removing the per-plane rearrangement (*SS3D_noP*), removing the per-height rearrangement (*SS3D_noH*), and eliminating the selective scanning strategy for 3-D feature sequences, instead using a weighted pooling of BEV features from different heights followed by modeling 2-D contextual information with *OSSM* as described in [13]. For the dense BEV contrastive learning module, in the variant

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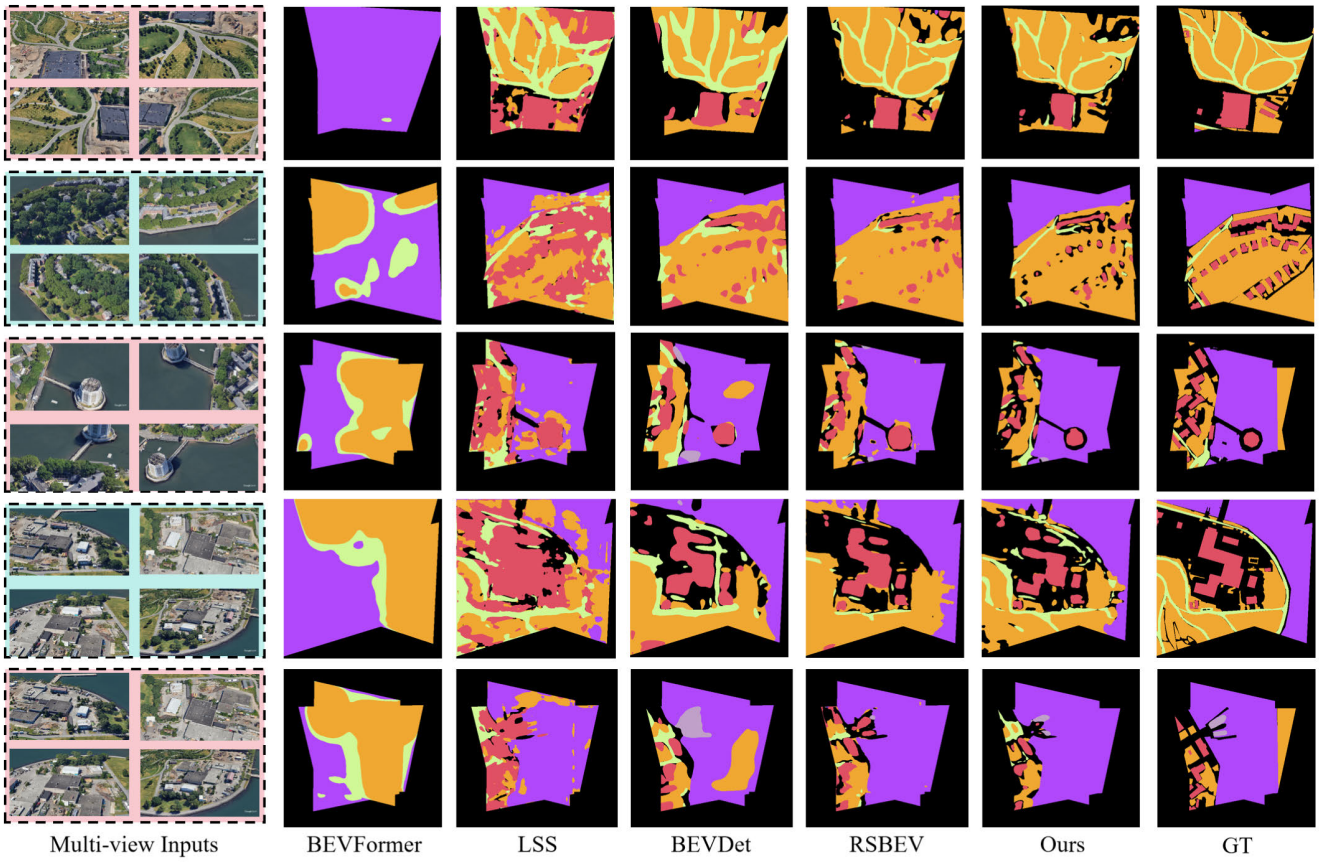



Fig. 6. Visualization comparison of BEVFormer [25], LSS [22], BEVDet [23], RSBEV [1], and our results. We present the BEV segmentation results from a series of multiview inputs on the left. Vegetation is shown in orange, buildings in red, roads in green, ships in light purple, water in dark purple, and other categories or occluded regions in black.

TABLE II
PERFORMANCE COMPARISON ACROSS REGIONS COVERED BY DIFFERENT NUMBERS OF CAMERAS

| Number of Cameras | Percentage of Area | IoU_vegetation | IoU_building | IoU_road | IoU_ship | IoU_water | mIoU↑ | F1 ↑ |
|---|---|---|---|---|---|---|---|---|
| 1 | 15.6% | 0.4285 | 0.5821 | 0.1743 | 0.8754 | 0.7890 | 0.5998 | 0.5838 |
| 2 | 35.8% | 0.5450 | 0.6221 | 0.5731 | 0.8763 | 0.8740 | 0.7260 | 0.6004 |
| 3 | 25.4% | 0.7513 | 0.6936 | 0.6827 | 0.8797 | 0.8828 | 0.7858 | 0.6535 |
| 4 | 23.2% | 0.7927 | 0.7004 | 0.6941 | / | 0.8853 | 0.7923 | 0.6517 |

*noCLM*, we simply omitted this component during the training process.

Table III shows the results from ablation studies, which underscore the robustness of our 3-D VMamba block and dense BEV contrastive learning module configurations. These results affirm the essential roles that each component plays in our model's architecture. Particularly, the outcomes associated with the OSSM model highlight the superior ability of our SS3D to capture and process complex spatial contexts effectively, which is pivotal for understanding intricate environmental layouts. Comparisons between SS3D_noH, SS3D_noP, and our comprehensive model configuration reveal that our tailored scanning strategies significantly enhance global spatial context integration within the 3-D domain. This integration is crucial especially for achieving high accuracy in multiview collaborative perception tasks such as semantic segmentation.

The differences between the noCLM variant and our complete method further illustrate the importance of our contrastive learning module. This module significantly improves feature alignment with GT, which is critical for maintaining the integrity of spatial relationships during BEV projections.

TABLE III
ABLATION STUDIES OF 3-D VMamba BLOCK AND DENSE BEV CONTRASTIVE LEARNING MODULE CONFIGURATIONS

| Model | mIoU_train | mIoU_val | F1_train | F1_val |
|---|---|---|---|---|
| OSSM | 0.9167 | 0.6953 | 0.7560 | 0.5981 |
| SS3D_noP | 0.9221 | 0.7074 | 0.7558 | 0.6055 |
| SS3D_noH | 0.9186 | 0.7298 | 0.7592 | 0.6210 |
| noCLM | 0.9314 | 0.7315 | 0.7630 | 0.6218 |
| Ours | 0.9195 | 0.7368 | 0.7563 | 0.6232 |

Our model not only consistently outperforms these variants in terms of mIoU and F1-scores, but also demonstrates the practical benefits of our unique approach in handling complex spatial information, making it highly effective for detailed environmental perception required in remote sensing applications.

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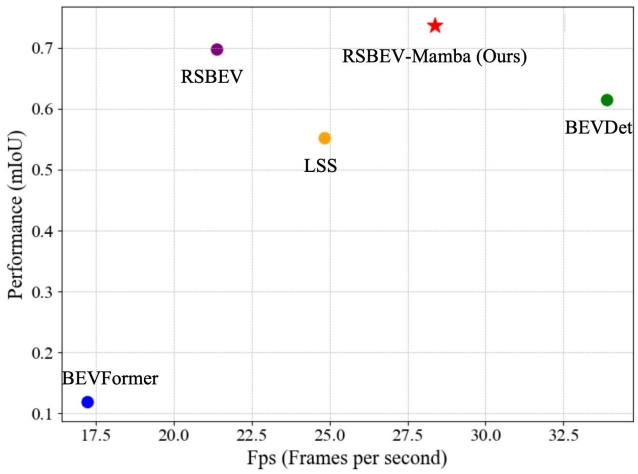



Fig. 7. Segmentation performance (mIoU) versus inference speed (frames/s) of five methods: BEVFormer, LSS, BEVDet, RSBEV, and our method (Ours). Each method is represented by a solid circle, with Ours marked by a star.

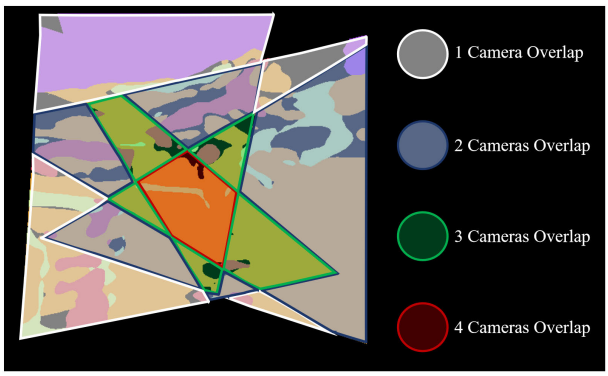



Fig. 8. BEV segmentation result with camera coverage markings. Based on the poses of four side-view cameras, the corresponding visible area in the BEV space for each camera is determined. The central region typically has a higher number of overlapping camera views. The color visualization method used is the same as in Fig. 6.

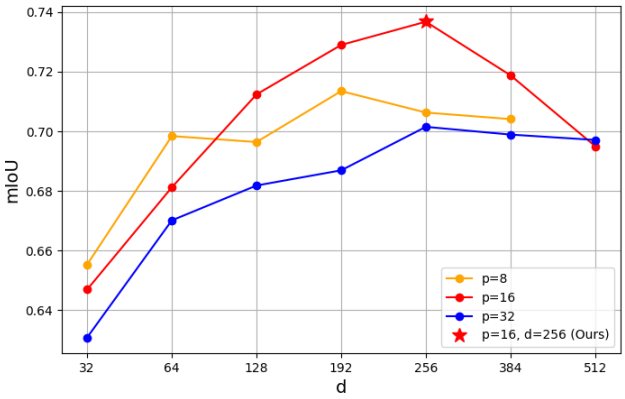



Fig. 9. Impact of varying patch sizes ($p$) and embedding dimensions ($d$) on mIoU. The plot shows the effect of different configurations on model performance, with p = 8 and d = 512 reaching GPU memory limitations, resulting in missing data points.

### E. Parameter Analysis

In this section, we explore the impact of the patch size ($p$) and embedding dimension ($d$) at the encoder input, along with the distribution of 3-D VMamba blocks across different stages of the encoder, on model performance. Patch size determines how input data are subdivided into manageable segments, influencing the granularity of the features processed, while the embedding dimension defines the size of the feature vectors, impacting the model's capacity to capture and process contextual information. Additionally, to validate the model's linear complexity, we tested GPU memory usage by varying the resolution of the BEV space with $p = 16$ and $d = 32$ held constant. Furthermore, through experimentation, we determined the optimal value of the coefficient $\lambda$ for the contrastive loss in the loss function.

TABLE IV
PERFORMANCE COMPARISON ACROSS DIFFERENT CONFIGURATIONS OF THE 3-D VMamba BLOCKS WITHIN OUR ENCODER

| stage1 | stage2 | stage3 | mIoU | F1 |
|---|---|---|---|---|
| 2 | 2 | 2 | 0.7217 | 0.6149 |
| 3 | 3 | 3 | **0.7330** | **0.6217** |
| 4 | 4 | 4 | 0.7294 | 0.6206 |
| 4 | 3 | 3 | 0.7241 | 0.6083 |
| 3 | 3 | 4 | 0.7280 | 0.6164 |
| 3 | 4 | 3 | **0.7324** | **0.6201** |
| 3 | 5 | 3 | 0.7339 | **0.6248** |
| 3 | 6 | 3 | **0.7368** | 0.6232 |
| 3 | 7 | 3 | 0.7351 | 0.6215 |
| 4 | 5 | 4 | 0.7298 | 0.6121 |

For the patch size, we tested patch sizes of $p \in \{8, 16, 32\}$ and embedding dimensions of $d \in \{32, 64, 128, 192, 256, 384, 512\}$. Fig. 9 illustrates the variation in mIoU with changes in these parameters. We observed consistent improvements in inference performance as $d$ increases, peaking at $d = 256$, beyond which the gains begin to plateau. The effect of patch size on performance was nuanced: extremely low (8) or high (32) values diminished effectiveness, likely due to insufficient global context capture or excessive information loss, respectively. The optimal balance was found at $p = 16$, where the model effectively managed local details without compromising the ability to model broader spatial relationships. Based on these findings, we configure our model with $p = 16$ and $d = 256$ to optimize both local detail preservation and global context understanding.

In our investigation into encoder configurations, as detailed in Table IV, we examined the influence of varying the number of 3-D VMamba blocks across different encoder stages. The results demonstrated that enhancements in model performance, as reflected in both mIoU and F1-score, are more pronounced when increasing the number of blocks in the middle stages of the encoder. This effect tends to saturate later compared to adjustments made in the initial and final stages of the encoder, suggesting a nuanced dynamic in how these blocks contribute to the network's overall efficacy. Our findings support the adoption of a spindle-shaped configuration for the distribution of 3-D VMamba blocks, where the middle stages of the encoder are equipped with a higher number of blocks compared to the ends. This configuration appears to be optimal for leveraging the capabilities of VMamba blocks to model complex spatial relationships effectively, leading to higher overall performance.

In the experimental testing model complexity, we utilized a Windows system with an NVIDIA RTX 4080 GPU (16 GB)

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TABLE V
PERFORMANCE COMPARISON ACROSS DIFFERENT VALUES OF $\lambda$ AFTER TRAINING FOR 5, 10, AND 50 EPOCHS

| Training Epochs | $\lambda = 0.1$ | $\lambda = 0.2$ | $\lambda = 0.3$ | $\lambda = 0.4$ | $\lambda = 0.5$ | $\lambda = 1$ |
|---|---|---|---|---|---|---|
| 5 | 0.7090 | 0.7136 | 0.7125 | 0.7144 | 0.7147 | 0.7195 |
| 10 | 0.7260 | 0.7238 | 0.7269 | 0.7274 | 0.7272 | 0.7290 |
| 50 | 0.7315 | 0.7344 | 0.7351 | 0.7298 | 0.7306 | 0.7286 |

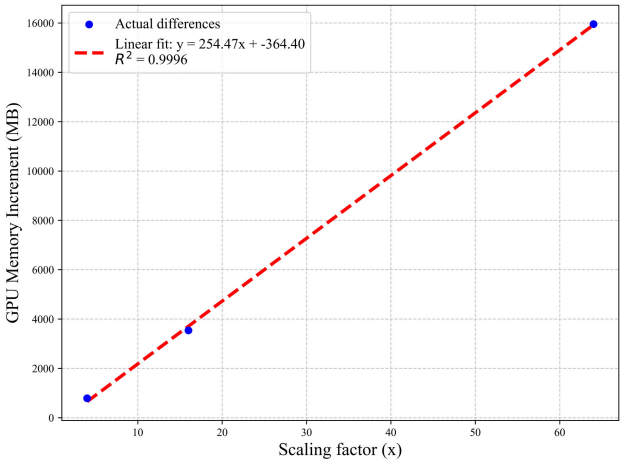



Fig. 10. GPU memory usage as the input sequence length scaling factor increases.

and an additional 24 GB of shared CPU memory due to the high memory requirements. The results of the complexity analysis are presented in Fig. 1(c). By setting the input resolution of $16 \times 16 \times 6$ as the baseline and calculating the incremental GPU memory usage as the input resolution doubles (corresponding to a $4\times$ increase in sequence length), we obtained the scatterplot shown in Fig. 10. The fit line achieves an $R^2 = 0.9996$. It demonstrates an excellent match to linear memory usage as the input sequence length increases, aligning with theoretical expectations. This linearity ensures that the model maintains acceptable memory usage even with 3-D feature inputs or larger BEV feature maps.

We tested different values of $\lambda$ (0.1, 0.2, 0.3, 0.4, 0.5, and 1) and evaluated the model's performance (mIoU) on the validation set after training for 5, 10, and 50 epochs. As shown in Table V, while a larger value of $\lambda$ results in faster convergence, it leads to lower performance after convergence. On the other hand, a smaller value of $\lambda$ slows down the convergence. Ultimately, we found that $\lambda = 0.3$ yields the best performance after 50 epochs.

## V. CONCLUSION AND FUTURE WORK

In this article, we introduced RSBEV-Mamba, a novel framework for multiview collaborative perception in remote sensing. By re-examining the modeling principles of BEV representation, we addressed the inherent depth uncertainty present in purely visual methods, enhancing the accuracy of view fusion from multiview image sets. The proposed framework integrates a 3-D BEV module that projects multiview 2-D image features into 3-D world coordinates, a 3-D VMamba module designed for efficient processing of densely projected features with linear computational complexity, and a dense BEV contrastive learning module that aligns BEV and GT features within the same dimensional framework using contrastive learning with the CLIP model. This configuration effectively leverages previously ignored projections at different heights, transforming them into valid 3-D sequences and enriching the contextual depth and precision of BEV encoding.

Meanwhile, this significant performance gain establishes the 3-D VMamba module as a versatile and efficient framework for 3-D perception tasks. Its sequence modeling approach enables the encoder to capture comprehensive spatial contextual information while maintaining linear computational complexity, making it well-suited for processing high-dimensional 3-D data and addressing a variety of applications requiring precise spatial understanding.

In future work, we plan to extend the RSBEV-Mamba framework to tasks like 3-D object detection and instance segmentation. However, the lack of publicly available datasets for multiview remote sensing currently limits experimental validation. We aim to address this by developing relevant datasets, contributing to the advancement of multiview remote sensing perception across diverse application domains.

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